4

BODY MEASUREMENTS

4.1 INTRODUCTION

In order to make a garment, it is essential to take the measurement of the person for whom the garment is being made. If the measurements are not taken properly then the garment will have an ill-fit. There are some rules of measuring and many ways of doing it. But we will first find out why it is necessary to measure correctly, the precautions to be observed while taking measurements, etc.

4.2 OBJECTIVE

After reading this lesson, you will be able to do the following:

- use the inchtape correctly;
- use the various methods of measuring;
- observe precautions during measuring;
- take measurements systematically.

4.3 USING THE INCH-TAPE FOR MEASURING

The tape which is used to take measurements is known as the "inch-tape". This is generally made of cloth or plastic. It has a metal strip of 3" at one end and another of ½" at the other end. The end with the 3" strip is used for measuring lengths and the side with the smaller strip is used for measuring widths and circumferences. The tape is marked in inches on one side and both in centimeter and inches on the reverse. Every inch is marked in 8 points.

So that precise measurements can be taken. In this way, the measuring tapes can be used to measure any part of the body for the garment to be made.

4.4 MEASURING CORRECTLY

The fit of a garment depends upon the cutting and stitching of the garment. The basis of cutting is measurement because only when proper measurements have been taken can a garment be cut properly. Hence, it becomes essential for a beginner to know the various methods of taking measurements so that a well fitting garment can be prepared. There are two methods of measuring:

i) Chest measuring system

ii) Direct measuring system

i) Chest Measuring System

In this system the chest measurement forms the basis of all other measurements used for the drafting.

ii) Direct Measuring System

In this system, the different parts of the body are precisely measured in order to get a well fitting garment.

Observe the following while measuring:

a) The inch-tape should not be bent or curled at any place.

b) While measuring circumference, place 2-3 fingers inside, below the tape and the thumb on the outside.

c) Use the ½" strip end to measure widths and circumferences and the 3" strip side for lengths.

There are basically three kinds of measures :-

1. **Lengths** - full length of the garment, shoulder to waist, length of sleeve, etc. and for lower garments - the length from waist down, crotch length, etc.

2. **Widths** - across shoulders, across back, etc.

3. **Circumferences** - The inch tape is wrapped round the area to be measured - round chest, round sleeve, trouser bottoms, round knee, hip, round waist, round neck, etc.

The human body has been divided into eight measurement zones in order to facilitate measurements. This is also known as the **'eight head principle'**. These zones are :



Fig 4.1 Measurement zones of the body (front view)

1. Top of the head to chin
2. Chin to nipple
3. Nipple to navel
4. Navel to pelvic arc - area below the hip.
5. Pelvic arc to mid thigh
6. Mid thigh to small of leg - narrow area below the knee
7. Small of leg to lower leg - narrow area below calves
8. Lower leg to heel

The eight head division is applicable to the height of an adult standing on the toes. Adequate knowledge of these eight zones of the body is very helpful in obtaining correct measurements for the garment to be made.

4.5 PRECAUTIONS WHILE TAKING MEASUREMENTS

The following precautions should be observed while measuring anyone:-

1. Stand to the right hand side of the person being measured.
2. Exercise strict self control as a tailor.
3. Talk only as much as required.
4. Be polite and simple in behaviour.
5. Avoid standing in front of the person being measured.
6. Some people are shy while being measured. Put them at ease by asking about the design of the garment, etc. while measuring them.
7. Take measurements systematically and sequentially.

8. Avoid touching the other person unnecessarily.
9. Be careful to see that the tape does not turn or fold at any place while measuring.
10. Use the end with the 3" metal strip to measure lengths and the other end to measure widths and circumferences.
11. Ask the other person to remove bulky extra garments like woolens, while taking measurements.
12. While measuring circumferences place two - three fingers inside, under the tape and the thumb on the outside.
13. Note down the measurements simultaneously in a notebook.
14. Observe the body for special adjustment needs while measuring.

INTEXT QUESTIONS 4.1

Fill in the blanks:

1. Stand to the ..... of the person being measured.
2. Measurements are taken by ..... methods, ... and .....
3. There are ..... main kinds of measurements.
4. The end with the 3" metal strip is used for measuring .....
5. The human body can be divided into ..... measurement zones.

ACTIVITY

Take measurements according to steps detailed earlier.

4.6 USE OF 'L' SCALE

This is made of wood, plastic or metal. It is shaped like the letter 'L' of English hence it is called the 'L' scale. It is 24" long on one side and 12" wide on the other side. It is graduated in inches. It is generally used by tailors as it facilitates their work and makes it easier. It can be used to mark vertical and horizontal lines simultaneously. It is used to make draftings. Repeated user of the inch-tape can be avoided by using the 'L' scale while drafting. The metal 'L' scale is used in factories for drafting on boards, cloth or paper.

4.7 SEQUENCE OF TAKING MEASUREMENT

Measurements must always be taken in a sequence. Be careful to measure all lengths if lengths are being measured or if widths are being measured, then measure all widths at

the same time. Taking measurements randomly, that is, one length and then one width and then back to a length again is not correct. Hence, one should know the sequence in which measurements must be taken.

1. ROUND MEASURES

i) Round Chest

This measure is called 'round bust' for ladies and 'round chest' for men. In order to take the correct measurement, pass the tape under the arms, across the back and hold the ends in front, in center of the chest.

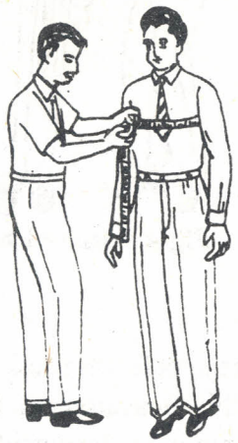


Fig 4.2

ii) Round Waist

This is also a round measure. The place to be measured is the narrow part below the chest and above the hips.

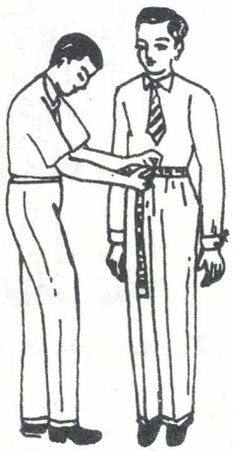


Fig 4.4 Round Waist

iii) Round hip

The protruding, extended part on the backside, is the hip area. The round measurement of the area is called the round hip. It is called 'hip' for women and 'seat' for men. It is advisable to insert two fingers below the tape while measuring.



Fig 4.4 Round hip

iv) Round neck

This is the round measurement for the neck. The end of tape with the smaller strip should be used to take this measurement. It is best sto keep one finger inside, under the tape and the thumb on the outside.



Fig 4.5 Round neck

v) Knee measurement

The measurement is taken according to the liking of the person. This is generally

taken for making tight fitting lower body garments, e.g. churidar pyjama, etc. The tape is passed under the knee and brought up on the top of the knee. While taking this measurement, the person should ideally be asked to sit on a chair with the knee bent at right angle.



Fig 4.6 Knee measurement

vi) Calf

The muscular portion of the leg below the knee is called the calf. This measurement is taken for well-fitting lower garments like churidar pyjama, etc.

vii) Round bottom

This measurement is taken both for upper and lower body garments. In the upper body garments it is called the round bottom sleeve and in lower body garments it is called lower round bottom, e.g. in trousers, pajamas, etc. This measurement varies with the prevailing fashion.



Fig 4.7 Round bottom

2. WIDTH MEASURES

i) Shoulders

This is to be taken from the backside. The measurement is from one end of the shoulders to the other.

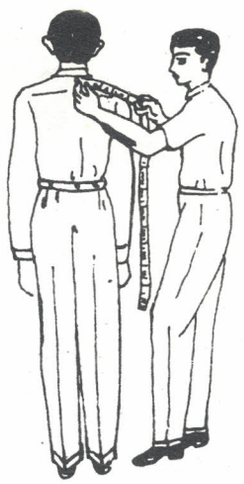


Fig 4.8 Shoulder measurement

ii) Across chest

This is to be taken from the front. It is measured at the narrowest part of the chest- from the middle of arm syce of one side to the other.

3. LENGTH MEASURES

i) Sleeve length

The end of the tape is placed at the top of the shoulder and the desired sleeve length is then measured.



Fig 4.9 Sleeve length.

ii) Shoulder to waist

This measurement is taken from the backside by placing the end of the taps at the base of the neck and measuring down to the waist which is the narrow part below the chest.



Fig 4.10 Shoulder to waist measurement

iii) Full length

The full length for upper body garments is taken from the front by placing the end of the tape at the base of neck and measuring upto the desired length of garment.



Fig 4.11 Full length measurement

iv) Length

The length is measured for lower body garments like trousers, etc, by placing the end of the tape one inch below the navel and measuring upto the toe. This length can be varied as desired.



Fig 4.12 Length measurement for trousers.

v) Crotch length

This length is measured from the point where both legs meet and is taken on the inside of the leg.



Fig 4.13 Crotch measurement

4.8 WHAT YOU HAVE LEARNT

In this lesson you have learnt how to measure correctly, how to use the measuring tapE the precautions to be observed while taking measurements and the sequence of taking measurements.

4.9 TERMINAL EXERCISE

1. Why should measurements be taken in a sequence? What is the sequence of taking measurements? Describe in detail.

2. Describe where the 'L' scale is used.

ACTIVITY

Differentiate between:

i) Chest measuring system and direct measuring system

ii) Inch tape and 'L' scale

4.10 ANSWERS TO INTEXT QUESTIONS

4.1 i) right ii) two, chest, direct

iii) three iv) widths, circumferences

v) eight.

MEASUREMENTS FOR CHILDREN

Age	Chest	Waist	Hip	Across Back	Neck	Sleeve	Natural Waist (Shoulder to waist)	Length Short	Length Frock	Length
1 Year	18" 46cm	18" 46cm	18" 46cm	8" 20cm	9" 23cm	10" 25cm	7" 18cm	9" 23cm	15"-16" 38-41cm	16"-18" 41-46cm
1-2	20" 51cm	20" 51cm	20" 51cm	8½ 22cm	9½ 25cm	11" 28cm	7½ 19cm	10" 25cm	16"-18" 41-46cm	18"-20 46-51cm
3-4	22" 56cm	22" 56cm	22" 56cm	9" 23cm	10" 25cm	12" 31cm	8" 21cm	11" 28cm	20"-22" 51-56cm	20"-22" 51-56cm
5-6	24" 61cm	22" 56cm	24" 61cm	10" 25cm	11" 28cm	14" 36cm	9" 23cm	12" 31cm	24" 60cm	22"-24" 56-61cm
7-8	26" 66cm	23" 58cm	28" 71cm	11" 28cm	11½ 29cm	17" 43cm	10" 25cm	13" 33cm	26" 66cm	26"-28" 66-71cm
9-10	28" 71cm	24" 61cm	30" 79cm	12" 31cm	12½" 32cm	19" 48cm	11½" 29cm	14½" 37cm	28" 71cm	28"-30" 71-76cm
11-12	30" 76cm	25" 64cm	33" 84cm	13" 33cm	13" 33cm	21" 54cm	13" 33cm	16" 41cm	30" 77cm	30"-32" 76"-81"

MEASUREMENTS FOR ADOLESCENTS AND LADIES

Age	Bust (Breast)	Waist	Hip	Across Back	Neck	Sleeve Length	Natural Length	Frock Length	Blouse
13-14	30" 76cm	25" 64cm	33" 84cm	13" 33cm	13" 33cm	21" 54cm	13" 33cm	32"-34" 81-86cm	13½" 35cm
15-16	32" 81cm	26" 66cm	35" 89cm	13½ 35cm	13½" 35cm	22" 56cm	13½" 35cm	35" 89cm	3½" 35cm
17-18	33" 86cm	26½" 69cm	36" 94cm	13½" 36cm	13½" 36cm	22" 56cm	13½" 36cm	36" 94cm	14" 37cm
19-20	35" 86cm	27½" 69cm	38" 94cm	14" 36cm	14" 36cm	22½" 56cm	14½" 36cm	38" 94cm	15" 37cm
21-22	35" 88cm	27½" 70cm	38" 97cm	14" 36cm	14" 36cm	22½" 58cm	14½" 37cm	38" 97cm	15" 38cm
23-24	36" 91cm	28" 71cm	40" 102cm	14" 36cm	14" 36cm	23" 58cm	15" 38cm	40" 102cm	16" 41cm

MEASUREMENTS FOR GENTS

Age	Chest	Waist	Hip	Across Back	Yoke	Round Neck	Sleeve	Neck to Waist	Shirt Length	Bush Length	Short Length	Pant Length	Crotch Length
13-14	30" 76cm	27" 69cm	33" 84cm	6½ 17cm	15" 38cm	13½" 34cm	22" 56cm	14" 36cm"	28" 71cm	26" 66cm	15" 38cm	37" 94cm	27" 69cm
15-16	32" 81cm	28" 71cm	34" 86cm	6 3/4" 17cm	16" 41cm	13 3/4" 35cm	22½" 57cm	14½" 37cm	29" 74cm	27" 69cm	15½" 39cm	38" 97cm	28" 71cm
16-17	33" 84cm	29" 74cm	35" 89cm	7" 18cm	16½" 42cm	14" 36cm	23" 59cm	15" 38cm	30" 76cm	27½" 70cm	16" 41cm	39" 99cm	28½" 72cm
18-19	34" 86cm	30" 76cm	36" 91cm	7 1/4" 19cm	17" 43cm	14½" 37cm	23½" 60cm	15½" 39cm	30½" 76cm	28" 71cm	16½" 42cm	40" 102cm	29" 74cm
20-21	35" 89cm	31" 79cm	37" 94cm	7½" 19cm	17½" 45cm	14½" 37cm	24½" 61cm	16" 41cm	3" 79cm	29" 74cm	17" 43cm	41" 104cm	29½" 75cm
22-23	36" 91cm	32" 81cm	38" 97cm	8" 20cm	18" 46cm	15" 38cm	24" 61cm	16½" 42cm	32" 82cm	30" 76cm	17½" 45cm	42" 107cm	30" 76cm
23-24	37" 94cm	33" 84cm	39" 99cm	8" 20cm	18" 46cm	15" 38cm	25" 64cm	16½" 42cm	33" 85cm	31" 79cm	18" 46cm	42" 1087cm	30" 76cm